



Assignment 1: Matrices

Matrices, Operations on Matrices,
Determinant, System of Equations

MAT113 – Mathematics for Biosciences

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Questions

1. Write the **order** of each matrix and identify the elements a_{13} , a_{21} , and a_{22} :

$$A = \begin{pmatrix} 4 & -1 & 7 \\ 0 & 3 & 5 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 1 \\ 0 & -3 \\ 5 & 6 \end{pmatrix}, \quad C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

2. Find x, y, z if

$$\begin{pmatrix} 2x & y-1 \\ z+2 & 4 \end{pmatrix} = \begin{pmatrix} 6 & 5 \\ 8 & 4 \end{pmatrix}.$$

3. Identify the type of each matrix below (row, column, square, diagonal, scalar, identity, symmetric, skew-symmetric, upper triangular, lower triangular):

(a) $\begin{pmatrix} 3 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & 0 & 3 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 2 & 3 \end{pmatrix}$

(c) $\begin{pmatrix} 0 & 4 \\ -4 & 0 \end{pmatrix}$

(d) $\begin{pmatrix} 7 & 0 & 0 \\ 2 & 5 & 0 \\ 3 & 1 & 4 \end{pmatrix}$

(e) $\begin{pmatrix} 1 & 3 \\ 3 & 2 \end{pmatrix}$

(f) $\begin{pmatrix} 2 \\ -1 \\ 5 \end{pmatrix}$

4. Verify that $AI_2 = I_2A = A$, where

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

Explain in one sentence why I_2 is called the identity matrix.

5. If

$$A = \begin{pmatrix} 2 & -1 \\ 3 & 0 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 4 \\ -2 & 5 \end{pmatrix},$$

compute $A + B$, $A - B$, and $2A + 3B$.

6. Find the scalar k such that $kA = B$, where

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 3 & 6 \\ 9 & 12 \end{pmatrix}.$$

7. For

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 5 & 6 \\ 7 & 8 \end{pmatrix},$$

verify the identity $(AB)^T = B^T A^T$.

8. Compute the transpose of

$$A = \begin{pmatrix} 1 & -2 & 3 \\ 0 & 4 & 5 \end{pmatrix}$$

and state its order.

9. Compute AB and BA for

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ 1 & 3 \end{pmatrix}.$$

Is $AB = BA$?

10. Let

$$A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{pmatrix}_{2 \times 3}, \quad B = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2 & 1 \end{pmatrix}_{3 \times 2}.$$

Compute AB . Explain why BA is not of the same order as AB .

11. Let

$$A = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}, \quad B = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}.$$

Show that $AB \neq BA$.

12. Let $D = \text{diag}(2, 4, -1)$ and $A = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}$. Compute DA and AD . Describe in words what happens to the rows of A in DA and to the columns of A in AD .

13. Let $A = \begin{pmatrix} 3 & -1 \\ 2 & 1 \end{pmatrix}$. Compute: $\text{tr}(A)$ and $\det(A)$

14. Let $A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \end{pmatrix}$. Compute $A^2 + 3A - 4I$.

15. Let $A = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. Compute:

(a) A^2, A^3, A^4 .

(b) Verify $A^2 + I = O$ using the characteristic equation.

16. Let

$$A = \begin{pmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{pmatrix}.$$

(a) Verify $A^3 - 4A^2 + 5A - 2I = O$.

(b) Using above expression, express A^4 in terms of A^2, A , and I .

17. Compute the determinant of

$$A = \begin{pmatrix} 3 & -2 \\ 5 & 4 \end{pmatrix}.$$

18. Compute the determinant of

$$A = \begin{pmatrix} 2 & 1 & 3 \\ 0 & 4 & -1 \\ 1 & 2 & 5 \end{pmatrix}$$

by expanding along the first column. Verify your answer by expanding along the first row.

19. Evaluate the following determinants **without** fully expanding them (use properties of determinants):

(a) $\begin{pmatrix} 2 & 3 & 5 \\ 2 & 3 & 5 \\ 7 & 1 & 4 \end{pmatrix}$

(b) $\begin{pmatrix} 1 & 2 & 3 \\ 0 & 4 & 5 \\ 0 & 0 & 6 \end{pmatrix}$

(c) $\begin{pmatrix} 1 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -3 \end{pmatrix}$

20. Use **Cramer's rule** to solve the system:

$$3x + 2y = 13$$

$$x - y = 1$$

21. Use **Cramer's rule** to solve the system:

$$x + y + z = 6$$

$$x - y + z = 2$$

$$2x + y - z = 1$$

22. **(Forensic Calibration Problem)** A forensic chemist prepares a calibration mixture using three standard drug solutions with unknown concentrations c_1, c_2, c_3 (in mg/L). Three independent spectrophotometric measurements give the following equations:

$$2c_1 + c_2 + c_3 = 8$$

$$c_1 + 3c_2 + 2c_3 = 11$$

$$3c_1 + c_2 + 2c_3 = 13$$

Use **Cramer's rule** to determine the concentrations c_1, c_2, c_3 . Interpret your result in one sentence.

Answer Key

1. A has order 2×3 ; $a_{13} = 7$, $a_{21} = 0$, $a_{22} = 3$. B has order 3×2 . C has order 3×3 .
2. $x = 3$, $y = 6$, $z = 6$.
3. (a) Scalar matrix (also diagonal, square). (b) Row matrix. (c) Skew-symmetric. (d) Lower triangular. (e) Symmetric. (f) Column matrix.
4. Both products equal A . I_2 is called the identity matrix because it behaves like the number 1 in matrix multiplication: $AI = IA = A$.
5. $A + B = \begin{pmatrix} 3 & 3 \\ 1 & 5 \end{pmatrix}$, $A - B = \begin{pmatrix} 1 & -5 \\ 5 & -5 \end{pmatrix}$, $2A + 3B = \begin{pmatrix} 7 & 10 \\ 0 & 15 \end{pmatrix}$.
6. $k = 3$.
7. $AB = \begin{pmatrix} 19 & 22 \\ 43 & 50 \end{pmatrix}$, so $(AB)^T = \begin{pmatrix} 19 & 43 \\ 22 & 50 \end{pmatrix}$. Also $B^T A^T = \begin{pmatrix} 5 & 7 \\ 6 & 8 \end{pmatrix} \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix} = \begin{pmatrix} 19 & 43 \\ 22 & 50 \end{pmatrix}$. Both sides match.
8. $A^T = \begin{pmatrix} 1 & 0 \\ -2 & 4 \\ 3 & 5 \end{pmatrix}$, order 3×2 .
9. $AB = \begin{pmatrix} 4 & 6 \\ 10 & 12 \end{pmatrix}$, $BA = \begin{pmatrix} 2 & 4 \\ 10 & 14 \end{pmatrix}$. $AB \neq BA$.
10. $AB = \begin{pmatrix} 7 & 5 \\ 16 & 11 \end{pmatrix}$. BA is 3×3 because B is 3×2 and A is 2×3 , so BA has order 3×3 .
11. $AB = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}$, $BA = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}$. Hence $AB \neq BA$.
12. $DA = \begin{pmatrix} 2 & 4 & 6 \\ 16 & 20 & 24 \\ -7 & -8 & -9 \end{pmatrix}$, $AD = \begin{pmatrix} 2 & 8 & -3 \\ 8 & 20 & -6 \\ 14 & 32 & -9 \end{pmatrix}$. In DA , row i of A is multiplied by d_i . In AD , column j of A is multiplied by d_j .
13. $|A| = 3(4) - (-2)(5) = 12 + 10 = 22$.
14. $|A| = 31$ (both expansions give 31).
15. $M_{23} = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} = 3$; $C_{23} = (-1)^5 \cdot 3 = -3$.
16. (a) 0 (two identical rows). (b) 24 (upper triangular: $1 \cdot 4 \cdot 6$). (c) -15 (diagonal: $1 \cdot 5 \cdot (-3)$).
17. $x = 3$, $y = 2$.
18. $x = 1$, $y = 2$, $z = 3$.
19. $|A| = 0$, so non-trivial solutions exist. General solution: $(x, y, z) = t(-1, -1, 1)$, $t \in \mathbb{R}$.

20. $c_1 = 2$, $c_2 = 1$, $c_3 = 3$ mg/L. *Interpretation: The calibration mixture contains 2 mg/L of drug 1, 1 mg/L of drug 2, and 3 mg/L of drug 3.*

— End of Practice Set —